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Force caused by Potential: Electro-Magnetic Field

- static: localized source charge at rest, only spatial dependency, no time dependency
- stationary: source charge at uniform movement, steady current flow, space and time dependency
- dynamic: source charge at fast movement, radiation, space and time dependency
- relativistic: source charge at very speedy movement, radiation, space and time dependency

- rest localization vs. time-dependency
- slow vs. fast movement
- vacuum vs. matter
- local vs. global of $\mathcal{B}$, singularity vs. regularity of $\mathcal{E}$
- macro vs. micro
- far-field vs. near-field

Magnetic Field in Vacuum

$$\mathcal{B} = \nabla \times U_m := \text{rot}(A) \stackrel{\text{Matter}}{=} \mu_0 (\mathcal{H} + \mathcal{M}) \stackrel{\text{Vacuum}}{=} \frac{1}{c} \left(\frac{r}{|r|} \times \mathcal{E} \right) \stackrel{\text{Far Field}}{=} \frac{1}{2} \frac{1}{2\pi} \mu_0 \frac{k^2 e^{i(kr - \omega t)}}{r} \frac{r}{|r|} \times \left(c p_{0_{dipol}} + m_0 \right) \quad (1)$$

Magnetic Field in Matter

Excitation $\hat{=}$ Displacement

$$\text{Excitation:} \quad \mathcal{H} = \frac{1}{\mu_0} \mathcal{B} - \mathcal{M} = \frac{1}{\mu} \mathcal{B} \quad \text{Magnetisation:} \quad \mathcal{M} = \frac{\mu - \mu_0}{\mu \mu_0} \mathcal{B} = \frac{\mu - \mu_0}{\mu_0} \mathcal{H} \quad (2)$$

Electric Field in Vacuum

$$\mathcal{E} \stackrel{\text{conservativ}}{=} -\nabla U_e \stackrel{\text{dynamic}}{=} -\nabla U_e - \frac{\partial}{\partial t} U_m \stackrel{\text{fast moving charge}}{=} \frac{1}{2} \frac{1}{2\pi} \frac{1}{\varepsilon_0} \frac{1}{r} \frac{1}{\left(1 - \frac{r}{|r|} \cdot \frac{v}{c}\right)^3} q \left[\underbrace{\left(1 - \frac{v^2}{c^2}\right) \frac{\frac{r}{|r|} - \frac{v}{c}}{r}}_{\text{uniform movement}} + \underbrace{\frac{\frac{r}{|r|} \times \left[\left(\frac{r}{|r|} - \frac{v}{c}\right)\right]}{r}}_{\text{acceleration}} \right] \quad (3)$$

Electric Field in Matter

Displacement $\hat{=}$ Excitation (field change)

dielectricum = non-conducting material

electricum = conductor medium

$$\text{Excitation: } \mathcal{D} = \varepsilon_0 \mathcal{E} + \mathcal{P} \stackrel{\text{isotropic medium in weak fields}}{=} \varepsilon_0 \mathcal{E} + \varepsilon_0 \chi \mathcal{E} = \varepsilon_0 (1 + \chi) \mathcal{E} = \varepsilon_0 4\pi r^3 n \langle p_m \rangle = 4\pi r^3 n \langle \mathcal{E}_m \rangle \quad (4)$$

Abbreviations

Φ = electric potential A = magnetic potential $j = \nabla \times \mathcal{H}$ = elec. current density

$\mathcal{E}$ = Electric Field $\mathcal{B}$ = Magnetic Field $\nabla \cdot \mathcal{B} = 0$ $\mathcal{D}$ = Displacement density of dielectricum $\mathcal{M}$ = magnetization

$\mathcal{P}$ = Polarisation $\mathcal{H}$ = magn. excitation, magn. field strength ("magn. displacement")

n = spacial molecule density in volume

$\alpha = 4\pi r^3 = \text{constant}$ χ = elec. susceptibility of dielectricum $\kappa = \frac{\mu - \mu_0}{\mu_0} = \text{magn. susceptibility}$

$\langle p_m \rangle$ = spacial average of magn. dipol moment $\langle \mathcal{E}_m \rangle$ = spacial average of electric field

$\bar{\mathcal{E}}_m$ = time average of electric field μ = magn. permeability μ_0 = magn. constant ...

ε = dielectricity constant $\varepsilon_0 = \dots$